Day 4 - Oct 23: Leap Year, Excel References, First Git Repo

Swagat | Revision sheet | Pushed repo: Swagat-97/git-practice

1. KEY IDEAS

Python: even / odd / zero and leap year

- % gives the remainder. n % 2 == 0 means even, otherwise odd. In Python -3 % 2 is 1, so -3 is odd.
- Zero goes first: 0 % 2 == 0 is True, so with the even test first, 0 would print Even.
- In a condition 0, '' and empty lists count as False. Use n == 0 when you mean zero.
- Leap year, strictest first: divisible by 400 -> leap. Else by 100 -> not leap. Else by 4 -> leap. Else not leap.
- Test one value per branch: 2000, 1900, 2024, 2023. Test a guess before trusting it (400 does not fit 2020).

Excel: references and functions

- Relative reference stores a position (same row, one column left), not a fixed cell. Dragging moves it.
- $ locks only the part right after it: A1 nothing, A1 both, $A1 column, A$1 row.
- Ask: when I drag, should this part STAY (use $) or CHANGE (no $)?
- Grid formula: =$A2*B$1 (prices down column A, rates across row 1). Drag trick: count steps, move only the unlocked parts.
- COUNT = numbers only. COUNTA = everything non-empty. SUM, AVERAGE, MIN, MAX skip text and blanks inside a range. A blank is NOT a 0: 0 counts and pulls the average down.
- COUNTIF(range, ">=40"): quotes needed. With a cell: ">="&D1.
- SUMIF(check range, condition, sum range). SUMIFS(sum range, check range, condition, ...). Sum range moves to the FRONT in SUMIFS. Conditions are AND.

Git: first repo

- Three places: working folder -> staging area (git add) -> repository (git commit). git add does not save a snapshot.
- One-time identity: git config --global user.name and user.email (Colab forgets them each session).
- git init (once, right folder), git add file, git commit -m "message" (always -m), git status, git log --oneline.
- Push: git branch -M main, git remote add origin URL, git push -u origin main. Create the GitHub repo EMPTY, or the two histories clash and the push is rejected.
- Never paste a token in chat. Remove it from the remote URL after pushing, then revoke it. Next topic: git clone.

2. EXAMPLES

```
n = int(input())
if n == 0:
    print('Zero')
elif n % 2 == 0:
    print('Even')
else:
    print('Odd')
```

```
if year % 400 == 0:   print('Leap year')
elif year % 100 == 0: print('Not a leap year')
elif year % 4 == 0:   print('Leap year')
else:                 print('Not a leap year')
```

```
=A1*$D$1            GST rate locked in D1
=$A2*B$1            whole price x rate grid
=COUNTIF(B2:B6,">=40")
=SUMIF(A2:A6,"A",B2:B6)
=SUMIFS(B2:B6,A2:A6,"A",B2:B6,">=40")
```

```
!git config --global user.name "Name"
!git config --global user.email "mail"
!git init
!git add example.py
!git commit -m "First commit"
!git log --oneline
!git branch -M main
!git push -u origin main
```

3. MY MISTAKES AND THE FIX

- Leap year: gave the definition (366 days) instead of the rule -> a rule says WHICH years qualify.
- Guessed 400 without testing it -> 2020 % 400 is 20, so it rejects a real leap year. Test guesses on known cases.
- Put the 4 test before 400 and 100 -> 1900 prints Leap year (wrong) and the 100 test becomes dead code. Strictest first.
- Said 2024 passes 'the first case' -> it passes the THIRD test. Name the test, then the output.
- Even/odd/zero recall: wrote 'if num is 0 it's even or else odd' -> the real reason is 0 % 2 == 0, so zero must be tested before the even test.
- Excel: answered 'relative reference?' and '% symbol' -> the lock symbol is $; explain the reason, not only the label.
- Said column 'stays B' for B$1 -> a free column MOVES (B1 -> C1 -> D1); only the row is locked.
- Wrote A$2 for the price -> lock only what must stay: $A2. Then B$2 -> built it from two parts instead: =$A2*B$1.
- Said 'nothing locked' for $A2 -> the $ locks the column. (That time the $ did not show on my screen.)

- COUNT returns the number of rows; AVERAGE errors on text -> wrong. COUNT counts numbers only; AVERAGE skips text and blanks.
- Used =IF to count passes -> IF works one cell at a time. COUNTIF counts cells that meet a condition.
- Wrote COUNTIF(SUMIF(...)) -> two different jobs, never nested here. SUMIF(check, condition, sum) is the order. Forgot the = sign on two formulas.
- Git: said 'git add' saves a version -> commit does. A plain git commit opens an editor; use -m. Wrote git commit -"msg" and 'example.. py' (typos). Answered 'label?' for config -> name and email.
- git status: said 'not committed' for a staged file -> the precise word is staged (Changes to be committed).

4. HOW I IMPROVED

- Reference rules: from wrong ($ guess) to predicting =$A2*D$1 and =$A5*B$1 with correct reasons.
- SUMIFS: wrote the argument order correctly with two conditions on the first attempt after the SUMIF correction.
- Git: ran the full flow alone: 3 commits, branch rename, push, clean remote URL. Explained why the GitHub repo is created empty in my own words.

5. PRACTICE TO RETRY (no solutions)

- Even/odd/zero and leap year, from a blank file (Oct 24 REVISE). Test 0, 7, 10, -3 and 1900, 2000, 2024, 2023.
- Excel: 18% GST on 20 prices with one absolute reference. Price x rate grid with one formula. COUNTIF pass count. SUMIFS with two conditions.
- Git: new folder, git init, 3 commits, push to a new empty repo, then git clone it back.

6. RECALL QUESTIONS (answer without notes)

- Why must the zero test come before the even test?
- Give the four leap-year tests in order. What does 1900 print if the 4 test is first?
- What does each $ position lock: A1, $A1, A$1?
- What does =$A2*B$1 become in D2 and in B5?
- COUNT vs COUNTA. Blank vs 0 for AVERAGE.
- Write SUMIF and SUMIFS for 'class A'. Which one puts the sum range first?
- What does git add do, what does git commit do, why -m? Why is the GitHub repo created empty?

7. OPEN ITEMS

- SQL 5-question check (Oct 22): still open. Revoke the GitHub token. Recall log: done (see README).